

Preparation of Papers for AIAA Technical Journals

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Nomenclature

A	=	state matrix
B	=	input matrix
C	=	measurement/output matrix
F	=	closed-loop state matrix
J	=	quadratic performance cost (dimensionless)
K	=	state-feedback gain matrix
L	=	observer gain matrix
Q, R	=	LQR weighting matrices (controller)
Q_o, R_o	=	LQR weighting matrices (observer)
W	=	controllability matrix
W_o	=	observability matrix
d(t)	=	process disturbance
n(t)	=	measurement noise
m	=	nonlinear state vector
n	=	nonlinear input vector
u	=	control input (linearized)
x	=	deviation state (linearized about equilibrium)
$\hat{x}$	=	state estimate
y	=	measurement/output vector
w_x, w_y, w_z	=	body angular rates (rad s ⁻¹)
(y_i, z_i)	=	image-plane coordinates of star <i>i</i> (dimensionless)
α_i, δ_i	=	right ascension and declination of axis/star <i>i</i> (rad)
φ, θ, ψ	=	roll, pitch, and yaw angles (rad)
τ_i	=	reaction-wheel torque for wheel <i>i</i> (N m)
($\dot{\cdot}$)	=	time derivative

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I. Introduction

The purpose of this project is to design and implement a controller and an observer that enable a spacecraft to maintain a fixed orientation such that an astronaut traveling with constant velocity can re-enter the craft through a hatch. The orientation of the spacecraft is maintained relative to a distant constellation. The controller design must be capable of recovering from incoming space debris, and the observer design must be able to handle substantial noise in sensor measurements.

Similar work has been done in the field of observer design in spacecraft attitude control. Reference [1] details elements of star tracker observer design and attitude estimation. It emphasizes the application of specific algorithms to filter unreliable star position measurements. The authors highlight a specific algorithm called the Attitude-Independent Method (AIM) algorithm, through which they test attitude estimation based on simulated data.

Spacecraft attitude control is a major aspect of all space-faring projects. Aside from deep-space exploration missions, attitude control is important in properly orienting satellites in Earth orbit. Reference [2] presents attitude estimation algorithms designed for computational efficiency to enable projects that involve smaller satellites with limited computational ability. Because satellites support critical communication and navigation services and must maintain attitude control within less than one degree of error, accurate and reliable attitude estimation is vital to the continued growth of modern satellite platforms.

II. Theory

A. Controller

1. Equations of motion

This system's equations of motion are given in standard form, provided by Professor Bretl and Professor Chang:

$$\dot{\mathbf{m}} = f(\mathbf{m}, \mathbf{n}), \quad (1)$$

$$\begin{bmatrix} \dot{\psi} & \dot{\theta} & \dot{\phi} & \dot{w}_x & \dot{w}_y & \dot{w}_z \end{bmatrix}^T = f\left(\begin{bmatrix} \psi & \theta & \phi & w_x & w_y & w_z \end{bmatrix}^T, \begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 \end{bmatrix}^T\right). \quad (2)$$

These equations of motion also depend on the location of reaction wheels on the satellite. A tetrahedral layout was chosen because it can provide roughly equal torque in every direction, which should distribute the load to limit input saturation. The right ascension and declination angles of the reaction wheels are given in degrees:

$$(\alpha_i, \delta_i) \in \{(0^\circ, -90^\circ), (60^\circ, 19.471^\circ), (180^\circ, 19.471^\circ), (300^\circ, 19.471^\circ)\}. \quad (3)$$

A full definition of the equations of motion is given in the code.

2. State-space model

The equilibrium state for the dynamical system can be described with zero angular velocities, $w_{xe} = w_{ye} = w_{ze} = 0$, and zero angular position, $\phi_e = \theta_e = \psi_e = 0$. This corresponds to a position pointing at the center of the constellation with no rotational velocity. The equilibrium input is composed of zero torque input for all reaction wheels, $\tau_{1e} = \tau_{2e} = \tau_{3e} = \tau_{4e} = 0$, because the system is floating in space and there are therefore no external torques that must be counteracted.

The state-space model used to linearize this system is given by

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \\ \mathbf{x}\mathbf{m} - \mathbf{m}_e, \quad & \mathbf{u}\mathbf{n} - \mathbf{n}_e. \end{aligned} \tag{4}$$

The state transition matrix $\mathbf{A}$ and input matrix $\mathbf{B}$ are given by

$$\mathbf{A} = \left. \frac{\partial f}{\partial \mathbf{m}} \right|_{\mathbf{m}_e, \mathbf{n}_e}, \quad \mathbf{B} = \left. \frac{\partial f}{\partial \mathbf{n}} \right|_{\mathbf{m}_e, \mathbf{n}_e}. \tag{5}$$

3. Controllability

Next, we establish controllability of the linearized system. Showing controllability confirms all modes can be influenced by the inputs, ensuring a valid solution and sufficient control about the equilibrium point. The linear system (4) is controllable if and only if the controllability matrix

$$\mathbf{W} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \tag{6}$$

is full column rank. In this case, there are 6 states, so $\text{rank}(\mathbf{W}) = 6$.

B. Observer design

1. Sensor model

The positional data for a distant constellation are collected by onboard cameras relative to their field of view. The sensor model for the constellation consisting of 7 stars is given by

$$g(\psi, \theta, \phi) = \begin{bmatrix} y_1 & z_1 & y_2 & z_2 & \dots & y_7 & z_7 \end{bmatrix}^T. \tag{7}$$

The equilibrium positions are chosen to be

$$(y_i, z_i)_{i=1}^7 \approx [(0, 0), (0, 0.26), (0.26, 0.26), (0.26, -0.26), (-0.26, -0.26), (-0.26, 0.26), (0, -0.26)]. \quad (8)$$

The data collected by these sensors are subject to significant noise, which can cause major issues for the controller by obscuring the true system orientation. To address this, an observer must be implemented to provide the controller with more accurate state estimates. Extending the state-space model to incorporate measurements, we have

$$\mathbf{y} = \mathbf{C}\mathbf{x}, \quad (9)$$

where

$$\mathbf{C} = \left. \frac{\partial g}{\partial \mathbf{m}} \right|_{\mathbf{m}_e, \mathbf{n}_e}. \quad (10)$$

The full state-space model of the system with both controller and observer design is

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} - \mathbf{L}(\mathbf{C}\hat{\mathbf{x}} - \mathbf{y}), \quad (11)$$

subject to $\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}}$, where $\hat{\mathbf{x}}$ is the state estimate. The matrix $\mathbf{L}$ is the observer gain.

2. Observability

Similar to controllability, the sensor model we have implemented must be observable for it to be possible to design a controller in which the state estimate converges to the actual state. The system is observable if and only if the observability matrix

$$\mathbf{W}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \quad (12)$$

has rank n . In this case, $\text{rank}(\mathbf{W}_o) = 6$, which is equal to our number of states, so the system is observable.

3. LQR-Based Controller and Observer Design

The controller uses full-state feedback based on the state estimate,

$$\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}}, \quad (13)$$

where $\mathbf{K}$ is chosen by solving an LQR problem that minimizes

$$\int_0^\infty (\mathbf{x}(t)^\top \mathbf{Q} \mathbf{x}(t) + \mathbf{u}(t)^\top \mathbf{R} \mathbf{u}(t)), dt. \quad (14)$$

Diagonal entries of $\mathbf{Q}$ weight attitude and rate states, while diagonal entries of $\mathbf{R}$ weight reaction-wheel torques. These entries are tuned by trial and error in simulation to balance tight attitude regulation against actuator saturation and excessive control effort. With this feedback, the closed-loop matrix $\mathbf{F} = \mathbf{A} - \mathbf{BK}$ is required to have eigenvalues with negative real parts to ensure asymptotic stability.

Because the constellation measurements are noisy, a state observer is used to reconstruct the state from the outputs. The observer has the form

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} - \mathbf{L}(\mathbf{C}\hat{\mathbf{x}} - \mathbf{y}), \quad (15)$$

with $\mathbf{L}$ chosen so that the estimation error dynamics are fast and well damped. The gain $\mathbf{L}$ is computed using the dual LQR formulation with weighting matrices $\mathbf{Q}_o$ and $\mathbf{R}_o$. In this tuning, diagonal entries of $\mathbf{Q}_o$ encode how much measurement noise is expected for each sensor (larger values mean the observer trusts that sensor less), while diagonal entries of $\mathbf{R}_o$ encode expected process disturbances in each state (larger values lead to more aggressive correction of that state). As with $\mathbf{Q}$ and $\mathbf{R}$, $\mathbf{Q}_o$ and $\mathbf{R}_o$ are adjusted iteratively in simulation to achieve accurate, noise-robust state estimates that work well with the controller.

III. Experimental Methods

For our controller and observer design to be considered successful, the astronaut shall make a safe entry 80% of the time. To verify that the success criterion is met, PyBullet will be used to simulate the spacecraft. We will run a total of 200 simulations, 60 s each, with random initial conditions. Sensor noise and space debris disturbances are enabled. The data generated by this simulation will be imported into a Jupyter Notebook for analysis in Python. The general success of the controller/observer will be quantified with

$$[\% \text{ successful reentries}] = \frac{[\# \text{ successful reentries}]}{[\# \text{ simulations}]}. \quad (16)$$

With sensor noise and debris disturbances enabled, the observer shall achieve a 95th-percentile attitude estimation error of $\leq 3.5^\circ$ over $t \in [20, 60]$ s in $\geq 80\%$ of simulated trials. To verify success, we compute the instantaneous attitude estimation error (degrees) as $\sqrt{(\psi - \hat{\psi})^2 + (\theta - \hat{\theta})^2 + (\phi - \hat{\phi})^2}$ at each time sample; take the 95th percentile of these values over $[20, 60]$ s. The requirement is satisfied if at least 80% of trials have this percentile value $\leq 3.5^\circ$.

1. Results and Discussion

The controller designed in the theory section was implemented and tested in the PyBullet simulation environment. The primary objective of these experiments was to validate the performance of the LQR controller and observer and determine the extent to which it meets the success criteria. Of the 200 simulations conducted, the controller/observer was able to successfully catch the astronaut 64% of the time.

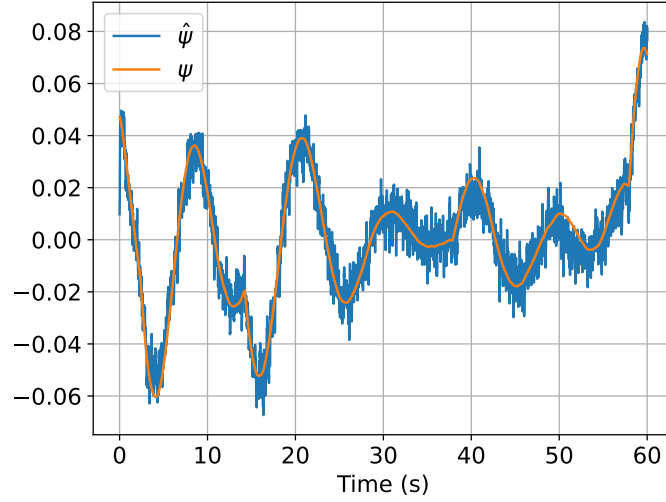


Fig. 1 This figure shows the state estimation for the yaw of the spacecraft in a given simulation trial.

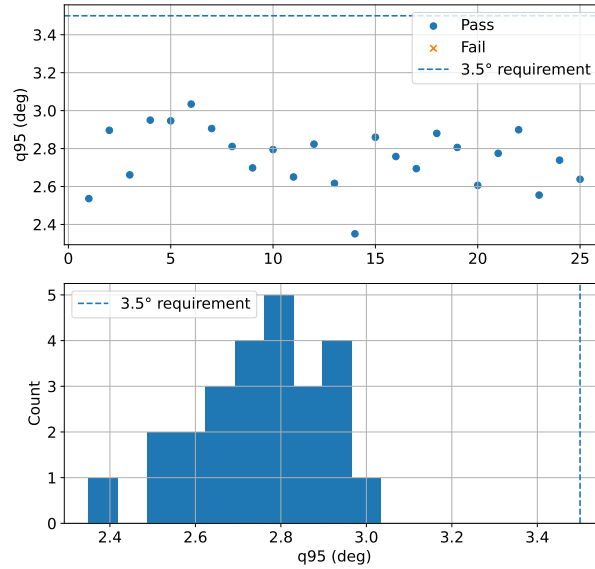


Fig. 2 This figure shows the 95th-percentile attitude estimation error for each trial, with all trials satisfying the 3.5° requirement.

The figures show that the observer consistently achieves small attitude errors across all trials, indicating a comfortable margin relative to the requirement and suggesting robust estimation performance under the tested disturbances and noise.

2. Conclusion

The objective of this project is to design a controller and observer for a satellite system which shall ensure safe entry for a returning astronaut. To design the controller and observer, we first designed a controller, and then designed an observer once an appropriate controller was selected. This meant that the controller was evaluated with no sensor noise and no observer intermediary (operating solely off of the actual state, rather than a state estimate). A set of seeds was chosen at random for simulations performed over some choice of controller, and the success rate of docking was used to select the best controller. Then, the same procedure was done for the observer design with noise enabled and feeding the state estimate to the controller. Overall, it was quite difficult to find a robust selection of both observer and controller design parameters. The best controller we designed had a success rate of roughly 85%, and the incorporation of a state estimate necessitated by sensor noise further exacerbated this difficulty. We found that a system with noise and external disturbances was difficult to reconcile, since the disturbances and the noise are difficult to interpret for the observer, and that large disturbances with small available torque made robust controller design infeasible for safety-critical applications.

Future work involving LQR controller/observer design can involve developing systematic ways to select the weight matrices. In this project, we largely relied on trial and error to solve this problem. One possible way to streamline the selection of weight matrices is the incorporation of direct observations or maximum limits into the components of the matrix. For example, future work can be done to formalize different ways of optimizing weight guesses based on what the controller limits, or desired convergence range is.

References

- [1] Delabie, T., Schutter, J. D., and Vandenbussche, B., “Robustness and Efficiency Improvements for Star Tracker Attitude Estimation,” *Journal of Guidance, Control, and Dynamics*, Vol. 38, No. 11, 2015, pp. 2108–2121. <https://doi.org/10.2514/1.G000894>, URL <https://doi.org/10.2514/1.G000894>.
- [2] Delabie, T., Schutter, J. D., and Vandenbussche, B., “Highly Efficient Attitude-Estimation Algorithm for Star Trackers Using Optimal Image Matching,” *Journal of Guidance, Control, and Dynamics*, Vol. 36, No. 6, 2013, pp. 1672–1680. <https://doi.org/10.2514/1.61082>, URL <https://doi.org/10.2514/1.61082>.